



MASTER IN ASTROPHYSICS
AND SPACE SCIENCE



TOR VERGATA
UNIVERSITÀ DEGLI STUDI DI ROMA

Computational Fluid Dynamics

INTRODUCTION TO NUMERICAL METHODS FOR ASTROPHYSICS

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Overview

Physical Background

Fundamental Definitions

Basic Equations

Examples

Numerical Approaches

Finite Difference Method

Finite Volume Method

Finite Element Method

Practical Implementation

Kelvin-Helmholtz Instability

Rayleigh-Taylor Instability

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Fluid Elements

- *Fluids*: things that flow, conventionally liquids and gases, contrast solids with rigid lattice
- *Elements*: region with definable local variables, fulfilling certain conditions
 - small enough to ignore systematic variations, size below any characteristic scale length

$$\ell \ll \left| \frac{q}{\nabla q} \right|$$

- large enough to ignore discreteness noise, statistically sufficient number of particles

$$n\ell^3 \gg 1$$

- in collisional fluids, large enough to communicate local conditions, requires interactions

$$\ell \gg \lambda$$

Describing Fluid Equations

- *Eulerian View*: cells of small volumes in fixed grid, flow is defined for all $\mathbf{t}$ at position $\mathbf{r}$ with rate of change $\partial/\partial\mathbf{t}$ for any measurable quantity Q
- *Lagrangian View*: follow particular fluid elements, comoving reference system with flow evaluated in each one via $D/D\mathbf{t}$ as the rate of change
- *Translating Relation*: add convective derivative term to account for movement of fluid element

$$\frac{DQ}{Dt} = \frac{\partial Q}{\partial t} + \mathbf{u} \cdot \nabla Q$$

Continuous Variables

■ *Primitive Fields:*

- velocity $\mathbf{u}$ (vector)
- density ρ (scalar)
- pressure p (scalar)
- temperature T (scalar)

■ *Conserved Quantities:*

- mass M (scalar)
- energy E (scalar)
- momentum $\mathbf{P}$ (vector)

■ *Derived Values:*

- viscosity ν (scalar)
- vorticity $\boldsymbol{\omega}$ (vector)

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Frequentist View

- *With Prior Knowledge*: out of all n different but equally likely possible outcomes of an event, there are k that have a property A which is therefore measured with probability

$$P(A) = \frac{k}{n}$$

- *Without Prior Knowledge*: out of n independent measurements the outcome A occurs k times, defining

$$P(A) = \lim_{n \rightarrow \infty} \frac{k}{n}$$

Bayesian View

- $P(A|B)$ quantifies the plausibility of an assumption A under known information given by B
- *Kolmogorov Axioms* underpin the rules for calculating with probabilities
- $P(A, B) = P(A) P(B|A)$ quantifies the probability of measuring A and B simultaneously, and from assuming $P(A, B) = P(B, A)$ we find

Bayes' Theorem

$$P(A|B) = \frac{P(B|A) P(A)}{P(B)}$$

Continuous Distributions

■ *Cumulative Distribution Function:* $F(x) = P(X \leq x)$

■ *Probability Density Function:* $f(x) = \frac{d}{dx} F(x)$

$$P(a \leq X \leq b) = \int_a^b f(x) dx = F(b) - F(a)$$

■ *Normalization:* $1 = \int_{-\infty}^{+\infty} f(x) dx$ $\lim_{x \rightarrow -\infty} F(x) = 0$ $\lim_{x \rightarrow +\infty} F(x) = 1$

Expectation Values

$$E[h(x)] = \int_{-\infty}^{+\infty} h(x) f(x) dx$$

■ *Mean* or first raw moment: $\mu = E[X] = \int_{-\infty}^{+\infty} x f(x) dx$

■ *Variance* or second raw moment: $\sigma^2 = E[(X - E[X])^2] = \int_{-\infty}^{+\infty} (x - E[x])^2 f(x) dx$

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■ Frameworks:

- Smoothed Particle Hydrodynamics (SPH)
- Lattice Boltzmann Method (LBM)
- Adaptive Mesh Refinement (AMR)

■ Regimes:

- *Plasma*: Magnetohydrodynamics (MHD)
- *Relativity*: Relativistic Hydrodynamics (RHD)
- *Degeneracy*: Quantum Effects
- *Collisionless*: Initial Distributions

■ Applications:

- Stellar Evolution
- Accretion Disks
- Cosmic Filaments

Thank You!

- C. Clarke & B. Carswell, *Principles of Astrophysical Fluid Dynamics* (Cambridge University Press) 2007
- L. A. Barba, *CFD Python: 12 Steps to Navier-Stokes*¹ 2022
- P. Mocz, *Medium*² 2020-2025

Resources on GitHub: [here](#)